
CF02: COMPLETE FORMALISM — CONTACT TO METRIPLECTIC EVOLUTION

A PREPRINT

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March 22, 2026

Abstract

We construct a *complete formalism* connecting *contact geometry* (the natural phase-space language for equilibrium thermodynamics and certain classes of dissipative systems) to *metriplectic* / *GENERIC*-style evolution laws of the form $\dot{q} = J \delta \mathcal{I} / \delta q + M \delta \Sigma / \delta q$, as used in the Void Dynamics Model (VDM) axioms. Starting from a contact manifold (M, α) with Reeb field R and (optionally) a compatible metric structure, we show how to: (i) obtain an antisymmetric Poisson operator J on constant-potential slices (symplectic leaves of $d\alpha$), (ii) select energy/invariant and entropy generators $(\mathcal{I}, \Sigma)$ consistent with thermodynamic interpretation, and (iii) construct a symmetric positive-semidefinite metric operator M using an information-geometric metric (Fisher–Rao / Ruppeiner) or a standard contact metric, together with an explicit projection that enforces the GENERIC degeneracy constraint $M \delta \mathcal{I} = 0$. The paper defines decisive falsifiers (Jacobi residual, symmetry/PSD of M , degeneracy residuals, and entropy production sign) and provides a fully specified algorithm plus a worked contact-metric example with numerical gate values.

Keywords: contact geometry; thermodynamics; metriplectic; GENERIC; Poisson bracket; information geometry; Void Dynamics Model

1 Objective and scope

This CF document provides a publication-grade bridge from *contact geometry* to *metriplectic* (GENERIC-style) evolution laws.

The target structure is the VDM two-generator split (A4–A5) [Lietz, 2025a]:

$$\dot{q} = J(q) \frac{\delta \mathcal{I}}{\delta q} + M(q) \frac{\delta \Sigma}{\delta q}, \quad (1)$$

with antisymmetry $J^\top = -J$, symmetry/positivity $M^\top = M \succeq 0$, and degeneracy constraints

$$J \frac{\delta \Sigma}{\delta q} = 0, \quad M \frac{\delta \mathcal{I}}{\delta q} = 0. \quad (2)$$

Contact thermodynamics supplies (i) a natural geometric container for the first law and Legendre structure [Mrugała, 1991, Bravetti et al., 2015], and (ii) a canonical “extra” direction generated by the Reeb field. Metriplectic/GENERIC supplies the complementary requirement: an *energy/invariant* conserved exactly while an *entropy* increases monotonically [Morrison, 1986, Grmela and Öttinger, 1997, Öttinger and Grmela, 1997, Öttinger, 2005].

2 Claim inventory and decisive falsifiers

Claim: C1 (Poisson sector from contact form) *Decisive metric: Jacobi residual of the induced Poisson tensor on a symplectic slice: $\|\text{Jacobi}(J)\| \leq \varepsilon$.*

Given a contact form α and a fixed-potential slice on which $\omega \equiv d\alpha$ has full rank, the induced bivector $J = \omega^{-1}$ defines a Poisson bracket on that slice.

Claim: C2 (Metric sector from contact metric / information metric) *Decisive metric: PSD and symmetry of M : $\lambda_{\min}(M) \geq -\varepsilon$, $\|M - M^\top\|_F \leq \varepsilon$.*

Given a symmetric positive-definite covariant metric g (e.g., Fisher–Rao/Ruppeiner on an equilibrium manifold [Weinhold, 1975, Ruppeiner, 1995, Amari and Nagaoka, 2000] or a standard contact metric [Geiges, 2008]), its inverse g^{-1} defines a symmetric positive-definite contravariant operator M_0 . A projection construction yields a degenerate M that preserves $\mathcal{I}$.

Claim: C3 (GENERIC degeneracy by construction) *Decisive metric: Degeneracy residuals: $\|J \delta \Sigma\| \leq \varepsilon$ and $\|M \delta \mathcal{I}\| \leq \varepsilon$.*

Choosing Σ compatible with the Poisson kernel (e.g., $\Sigma = s$ when J acts only on the symplectic coordinates) and applying the explicit projection $M = P_{\mathcal{I}} M_0 P_{\mathcal{I}}$ enforces degeneracy.

Claim: C4 (Entropy production sign) *Decisive metric: $\dot{\Sigma} = (\Sigma, \Sigma)_M \geq -\varepsilon$ on a validation domain..*

For symmetric $M \succeq 0$, the metric bracket yields nonnegative entropy production $\dot{\Sigma} = \langle \nabla \Sigma, M \nabla \Sigma \rangle \geq 0$.

3 Contact geometry essentials

3.1 Contact manifolds and Reeb splitting

Definition 1 (Contact manifold). A *contact manifold* is a $(2n+1)$ -dimensional smooth manifold M equipped with a one-form α such that

$$\alpha \wedge (d\alpha)^n \neq 0 \quad \text{everywhere on } M. \quad (3)$$

The hyperplane field $\xi \equiv \ker \alpha \subset TM$ is the *contact distribution*.¹

The *Reeb vector field* R is uniquely defined by [Geiges, 2008, de León and Lainz Valcázar, 2019]

$$\iota_R \alpha = 1, \quad \iota_R d\alpha = 0. \quad (4)$$

This yields a pointwise splitting

$$TM = \langle R \rangle \oplus \xi. \quad (5)$$

The two-form $d\alpha$ restricts to a nondegenerate (symplectic) form on ξ .

3.2 Darboux coordinates and a standard contact form

Locally (Darboux), one may choose coordinates (q^i, p_i, s) such that [Geiges, 2008]

$$\alpha = ds - p_i dq^i, \quad d\alpha = dq^i \wedge dp_i, \quad R = \frac{\partial}{\partial s}. \quad (6)$$

3.3 Legendre submanifolds and equilibrium thermodynamics

Definition 2 (Legendre submanifold). A *Legendre submanifold* $L \subset M$ is an n -dimensional submanifold tangent to ξ , i.e. $\alpha|_L = 0$. Equivalently, $TL \subset \xi$ and $d\alpha|_{TL} = 0$, so L is Lagrangian inside the symplectic vector bundle $(\xi, d\alpha|_\xi)$.²

¹Standard references include Geiges [2008].

²See Geiges [2008] for the contact-topology definition; in thermodynamics applications see Mrugała [1991], Bravetti et al. [2015].

In equilibrium thermodynamics, Legendre submanifolds encode equations of state: they are generated by thermodynamic potentials and Legendre transforms. For example, with coordinates (S, V, U, T, p) , the first law can be written as the vanishing of the pullback of the contact form

$$\alpha = dU - T dS - p dV. \quad (7)$$

4 Contact Hamiltonian systems

4.1 Contact Hamiltonian vector field

Given a smooth function $K : M \rightarrow \mathbb{R}$, the associated *contact Hamiltonian vector field* X_K is defined by the pair of conditions [de León and Lainz Valcázar, 2019, Bravetti, 2017]

$$\iota_{X_K} d\alpha = dK - (RK)\alpha, \quad \alpha(X_K) = -K. \quad (8)$$

In Darboux coordinates (q^i, p_i, s) with $R = \partial_s$, this gives the explicit formula

$$X_K = \frac{\partial K}{\partial p_i} \frac{\partial}{\partial q^i} - \left(\frac{\partial K}{\partial q^i} + p_i \frac{\partial K}{\partial s} \right) \frac{\partial}{\partial p_i} + \left(p_i \frac{\partial K}{\partial p_i} - K \right) \frac{\partial}{\partial s}. \quad (9)$$

Remark 1 (Common special case). If K is independent of s , then the (q, p) -subsystem reduces to ordinary symplectic Hamiltonian evolution with Hamiltonian $K(q, p)$, while the Reeb component tracks the potential coordinate via $\dot{s} = p_i K_{p_i} - K$.

4.2 Jacobi (contact) bracket

A contact manifold carries a canonical *Jacobi* bracket on functions [Geiges, 2008, de León and Lainz Valcázar, 2019]:

$$\{f, g\}_c = d\alpha(X_f, X_g) + f R(g) - g R(f). \quad (10)$$

This bracket is bilinear, antisymmetric, and satisfies the Jacobi identity, but it is not a Poisson bracket globally (it fails the strict Leibniz rule); Poisson structure appears on symplectic slices where s is fixed and R drops out.

5 Metriplectic / GENERIC structure

A metriplectic (GENERIC-style) evolution law is built from two brackets [Morrison, 1986, Grmela and Öttinger, 1997, Öttinger and Grmela, 1997, Öttinger, 2005]:

- an antisymmetric Poisson bracket $\{\cdot, \cdot\}$ induced by a Poisson operator J ,
- a symmetric positive-semidefinite metric bracket $(\cdot, \cdot)$ induced by a metric operator M .

For functionals A, B on the state space,

$$\{A, B\} = \left\langle \frac{\delta A}{\delta q}, J \frac{\delta B}{\delta q} \right\rangle, \quad (A, B) = \left\langle \frac{\delta A}{\delta q}, M \frac{\delta B}{\delta q} \right\rangle, \quad (11)$$

where $\langle \cdot, \cdot \rangle$ denotes the natural pairing (dot product in finite dimensions, L^2 pairing in field theory).

The VDM axioms identify $\mathcal{I}$ with an invariant (energy-like) functional and Σ with an entropy-like functional. The degeneracy constraints

$$J \frac{\delta \Sigma}{\delta q} = 0, \quad M \frac{\delta \mathcal{I}}{\delta q} = 0, \quad (12)$$

ensure:

$$\frac{d\mathcal{I}}{dt} = 0, \quad \frac{d\Sigma}{dt} = (\Sigma, \Sigma) \geq 0. \quad (13)$$

6 Construction: from contact data to a metriplectic split

6.1 Poisson operator from the contact form

On any slice where the restriction of $\omega = d\alpha$ is nondegenerate, define

$$J \equiv \omega^{-1}. \quad (14)$$

In Darboux coordinates on the (q, p) -subspace,

$$J_{(q,p)} = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}, \quad (15)$$

which satisfies antisymmetry and the Jacobi identity.

6.2 Metric operator from a thermodynamic / information metric

To obtain a symmetric bracket, choose a covariant metric g appropriate to the physical setting:

- **Information-geometry route:** Fisher–Rao metric on a statistical manifold and its thermodynamic reductions [Amari and Nagaoka, 2000, Bravetti et al., 2015].
- **Thermodynamic geometry route:** Weinhold/Ruppeiner Hessian metrics on equilibrium manifolds [Weinhold, 1975, Ruppeiner, 1995].
- **Purely geometric route:** a standard contact metric (e.g., $dq^2 + dp^2 + \alpha^2$ in Darboux charts) [Geiges, 2008].

Let $M_0 \equiv g^{-1}$ denote the associated contravariant operator. To enforce energy/invariant preservation, define the (pointwise) projector onto the orthogonal complement of $\nabla \mathcal{I}$:

$$P_{\mathcal{I}} = I - \frac{\nabla \mathcal{I} \nabla \mathcal{I}^\top}{\|\nabla \mathcal{I}\|^2}, \quad (16)$$

and set

$$M = P_{\mathcal{I}} M_0 P_{\mathcal{I}}. \quad (17)$$

If M_0 is symmetric PSD, then M is symmetric PSD, and satisfies $M \nabla \mathcal{I} = 0$ by construction.

6.3 Choosing Σ to satisfy $J \delta \Sigma = 0$

The remaining degeneracy constraint is model-dependent:

- If the state includes a contact-potential coordinate s and J acts only on (q, p) , then choosing $\Sigma = s$ makes $J \nabla \Sigma = 0$ identically.
- In field-theoretic Poisson systems (fluids/plasmas), nontrivial Casimirs exist; in such cases Σ may be chosen as a Casimir of J (or a function thereof), aligning with standard GENERIC practice [Morrison, 1986, Öttinger, 2005].

7 Worked example: a contact-metric metriplectic system

We demonstrate the construction in the simplest Darboux chart (q, p, s) with

$$\alpha = ds - p dq, \quad d\alpha = dq \wedge dp, \quad J = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (18)$$

Choose the invariant (energy) and entropy generators

$$\mathcal{I}(q, p, s) = \frac{1}{2} (p^2 + \omega^2 q^2), \quad \Sigma(q, p, s) = s. \quad (19)$$

7.1 Contact metric and raw operator M_0

Take the standard contact metric

$$g = dq^2 + dp^2 + \alpha^2 = dq^2 + dp^2 + (ds - p dq)^2, \quad (20)$$

whose covariant matrix in the (q, p, s) basis is

$$[g] = \begin{bmatrix} 1 + p^2 & 0 & -p \\ 0 & 1 & 0 \\ -p & 0 & 1 \end{bmatrix}. \quad (21)$$

This matrix has determinant 1 and inverse

$$[M_0] = [g]^{-1} = \begin{bmatrix} 1 & 0 & p \\ 0 & 1 & 0 \\ p & 0 & 1 + p^2 \end{bmatrix}, \quad (22)$$

which is symmetric and positive definite.

7.2 Projected M and gate values

We apply Equation (17) with $\omega = 2$ and validate over $q, p \in [-1, 1]$ on a 51×51 grid (with s fixed). The following gate values were computed using a deterministic numerical sweep (the reference implementation exists in `Derivation/Notebooks/00_Metriplectic/CF2_Contact_to_Metriplectic_Evolution.ipynb`):

Gate metric	Threshold	Observed (grid sweep)
Symmetry residual $\ M - M^\top\ _F$	$\leq 10^{-10}$	1.9×10^{-16}
PSD check $\lambda_{\min}(M)$	$\geq -10^{-10}$	-4.8×10^{-16}
Degeneracy $\ M\nabla\mathcal{I}\ $	$\leq 10^{-10}$	1.1×10^{-15}
Entropy production $\dot{\Sigma}$ minimum	$\geq -10^{-12}$	1.0

Table 1: Representative metriplectic gates for the contact-metric construction. Numerical errors are at machine precision.

Because $\Sigma = s$ and J has no s -components, the Poisson degeneracy $J\nabla\Sigma = 0$ holds identically.

8 Limitations and extensions

- **Contact bracket vs Poisson bracket:** the global contact/Jacobi bracket is not a Poisson bracket; the Poisson operator used in metriplectic form is obtained on symplectic leaves (fixed-potential slices) or via an extended Poisson structure on a subspace.
- **Choice of $\mathcal{I}, \Sigma$:** these generators are not uniquely fixed by contact geometry alone. Thermodynamic interpretation (which potential is used as s , which variables are extensive/intensive) matters [Mrugała, 1991, Bravetti et al., 2015].
- **Field-theory generalization:** in PDE settings, J and M are operators on function spaces; locality (finite propagation) and entropy-production gates must be verified numerically/analytically in the intended discretization [Lietz, 2025b].

9 Validation gates summary

The minimal publishable gate set for this CF is:

- **Poisson sector:** $J^\top = -J$ and Jacobi residual below tolerance.
- **Metric sector:** $M^\top = M$, $M \succeq 0$, and degeneracy $M\delta\mathcal{I} = 0$.
- **Entropy law:** $\dot{\Sigma} = (\Sigma, \Sigma)_M \geq 0$ on a domain of interest.

10 Provenance and reproducibility

Commit: 91332f4cf89a52d239eaab6659b8df8d8b9f3202 **Seed(s):** N/A **Artifacts:** **Source:**
 Derivation/Complete-Formalisms/CF02_Contact_to_Metriplectic_Evolution.md (SHA256:
 a7f5071514fa3c949a50e214a25bde68a46ec5226470a909f22532097e7e7091)

Reference implementations:

- Notebook: Derivation/Notebooks/00_Metriplectic/CF2_Contact_to_Metriplectic_Evolution.ipynb
- Canonical gates: 00_VALIDATION_METRICS.md [Lietz, 2025b]

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